

Nonlinear Optimization: Homework #3

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1 Convex Problems

1st Problem

57. PAPR. Consider a wireless transmitter with n antennas. The transmitter sends data in blocks of length T . A typical block is denoted $x := (x(1), x(2), \dots, x(T))$ with $x(t) \in \mathbf{R}^n$. To simplify, we are assuming that the random data vectors are real-valued; in practice, they would be complex-valued (in the baseband domain).

The peak-to-average power ratio (PAPR) of a block x is defined as

$$\text{PAPR}(x) = \frac{\max \{\|x(t)\|^2 : t = 1, \dots, T\}}{\mathbf{E} \left(\frac{1}{T} \sum_{t=1}^T \|x(t)\|^2 \right)}.$$

The PAPR assesses how unbalanced the power is spread across the block by computing the ratio between the peak and the average power. The symbol $\mathbf{E}$ denotes the expectation operator. (You don't need to know what $\mathbf{E}$ is or how to manipulate it. We are going to formulate the optimization problem for you. This is just context.)

From an electronics implementation viewpoint, it is desirable to have blocks with low PAPR. An approach consists in adding a “corrective” signal to each block; instead of transmitting the block x , we transmit the modified block $\hat{x} := (\hat{x}(1), \dots, \hat{x}(T))$ where

$$\hat{x}(t) = x(t) + S(t)w$$

for all $t = 1, \dots, T$. Here,

$$S(t) = \begin{bmatrix} s_1(t) & s_2(t) & \dots & s_p(t) \end{bmatrix} \in \mathbf{R}^{n \times p}$$

is a given set of available corrective signals to be used at time t , and $w = (w_1, w_2, \dots, w_p)$ is a vector of weights. In words, the corrected block $\hat{x}$ is obtained from x by adding a linear combination of the corrective signals. The goal of this problem is to determine the coefficients of the linear combination, i.e., the weight vector $w \in \mathbf{R}^p$.

Ideally, we would minimize $\text{PAPR}(\hat{x})$ with respect to w while also keeping the average power of the block $\hat{x}$ close to the original one (we don't want to spend too much extra power). However,

$$\begin{aligned} \text{PAPR}(\hat{x}) &= \frac{\max \{\|\hat{x}(t)\|^2 : t = 1, \dots, T\}}{\mathbf{E} \left(\frac{1}{T} \sum_{t=1}^T \|\hat{x}(t)\|^2 \right)} = \frac{\max \{\|x(t) + S(t)w\|^2 : t = 1, \dots, T\}}{\mathbf{E} \left(\frac{1}{T} \sum_{t=1}^T \|x(t) + S(t)w\|^2 \right)} \\ &= \frac{\max \{\|x(t) + S(t)w\|^2 : t = 1, \dots, T\}}{\mathbf{E} \left(\frac{1}{T} \sum_{t=1}^T \|x(t)\|^2 \right) + \frac{1}{T} w^T \sum_{t=1}^T S(t)^T S(t) w}. \end{aligned}$$

To obtain the last identity, we assumed the mean of the random data vectors $x(t)$ to be zero. Defining

$$P := \mathbf{E} \left(\frac{1}{T} \sum_{t=1}^T \|x(t)\|^2 \right) \quad \text{and} \quad R := \frac{1}{T} \sum_{t=1}^T S(t)^T S(t)$$

we have

$$\text{PAPR}(\hat{x}) = \frac{\max \{\|x(t) + S(t)w\|^2 : t = 1, \dots, T\}}{P + w^T R w}.$$

The function $\text{PAPR}(\hat{x})$ is difficult to minimize since it is nonconvex with respect to w (the constant $P > 0$ and the matrix R , assumed positive definite, are given). Instead, we formulate the convex problem

$$\underset{w}{\text{minimize}} \quad \max \{\|x(t) + S(t)w\|^2 : t = 1, \dots, T\} + \rho w^T R w.$$

where $\rho > 0$ is given. The idea is to minimize the peak power of $\hat{x}$ while trying to maintain its average power close to the original one by penalizing the injection of total extra power (the term $w^T R w$).

Express the problem as a SOCP or a SDP.

Solution to the 1st Problem. We apply an epigraph reformulation, introducing variables $\alpha, \beta \in \mathbf{R}$ that upper-bound the peak power and the total extra power respectively, so that the objective becomes a conic combination in the decision variables, that is,

$$\begin{aligned} & \underset{w, \alpha, \beta}{\text{minimize}} \quad \alpha + \rho\beta \\ & \text{subject to} \quad \max \{\|x(t) + S(t)w\|_2^2 : t = 1, \dots, T\} \leq \alpha, \\ & \quad \quad \quad w^T R w \leq \beta. \end{aligned} \tag{1}$$

Feasibility of these constraints already implies $\alpha \geq 0$ and $\beta \geq 0$, since the left-hand sides are squared norms or quadratic forms with $R > 0$. Then, the first constraint can be unfolded into T inequalities,

$$\max \{\|x(t) + S(t)w\|_2^2 : t = 1, \dots, T\} \leq \alpha \iff \|x(t) + S(t)w\|_2^2 \leq \alpha, \quad t = 1, \dots, T,$$

since each term is bounded by the maximum, achieved at some index. Substituting in (1) gives

$$\begin{aligned} & \underset{w, \alpha, \beta}{\text{minimize}} \quad \alpha + \rho\beta \\ & \text{subject to} \quad \|x(t) + S(t)w\|_2^2 \leq \alpha, \quad t = 1, \dots, T, \\ & \quad \quad \quad w^T R w \leq \beta. \end{aligned} \tag{2}$$

To match the form of the constraints, we rewrite $w^T R w$ as a squared ℓ_2 norm. Since $R > 0$, its eigenvalue decomposition $R = Q\Lambda Q^T$ exists, with Q orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$, where $\lambda_i > 0$ for $i = 1, \dots, p$. Its square root $R^{1/2} := Q\Lambda^{1/2}Q^T$ satisfies $(R^{1/2})^T = (Q\Lambda^{1/2}Q^T)^T = Q\Lambda^{1/2}Q^T = R^{1/2}$ and $R^{1/2}R^{1/2} = Q\Lambda^{1/2}(Q^T Q)\Lambda^{1/2}Q^T = Q\Lambda Q^T = R$, and so

$$w^T R w = w^T R^{1/2} R^{1/2} w = (R^{1/2} w)^T (R^{1/2} w) = \|R^{1/2} w\|_2^2.$$

Substituting, (2) becomes

$$\begin{aligned} & \underset{w, \alpha, \beta}{\text{minimize}} \quad \alpha + \rho\beta \\ & \text{subject to} \quad \|x(t) + S(t)w\|_2^2 \leq \alpha, \quad t = 1, \dots, T, \\ & \quad \quad \quad \|R^{1/2} w\|_2^2 \leq \beta. \end{aligned} \tag{3}$$

Next, we use the fact that, for $u \in \mathbf{R}^d$, $d \in \mathbf{Z}_+$, and $y, z \in \mathbf{R}$,

$$\begin{cases} \|u\|_2^2 \leq yz, \\ y \geq 0, \\ z \geq 0 \end{cases} \iff \left\| \begin{bmatrix} 2u \\ y - z \end{bmatrix} \right\|_2 \leq y + z \quad \left(\begin{array}{l} \text{Hyperbolic constraints as} \\ \text{second-order cone constraints} \end{array} \right). \tag{4}$$

Therefore, we reformulate the constraints in (3) as follows.

- Applying (4) with $u = x(t) + S(t)w \in \mathbf{R}^n$, $y = \alpha$ and $z = 1$, for each $t = 1, \dots, T$, gives

$$\|x(t) + S(t)w\|_2^2 \leq \alpha \iff \left\| \begin{bmatrix} 2(x(t) + S(t)w) \\ \alpha - 1 \end{bmatrix} \right\|_2 \leq \alpha + 1; \quad (5)$$

- Applying (4) with $u = R^{1/2}w \in \mathbf{R}^p$, $y = \beta$ and $z = 1$, gives

$$\|R^{1/2}w\|_2^2 \leq \beta \iff \left\| \begin{bmatrix} 2R^{1/2}w \\ \beta - 1 \end{bmatrix} \right\|_2 \leq \beta + 1. \quad (6)$$

Substituting these equivalent constraints gives

$$\begin{aligned} & \underset{w, \alpha, \beta}{\text{minimize}} && \alpha + \rho\beta \\ & \text{subject to} && \left\| \begin{bmatrix} 2(x(t) + S(t)w) \\ \alpha - 1 \end{bmatrix} \right\|_2 \leq \alpha + 1, \quad t = 1, \dots, T, \\ & && \left\| \begin{bmatrix} 2R^{1/2}w \\ \beta - 1 \end{bmatrix} \right\|_2 \leq \beta + 1. \end{aligned} \quad (7)$$

The problem in (7) is therefore formulated as a SOCP, as the objective is linear in the decision variables, and each constraint constrains an affine image of (w, α, β) to lie in a second-order cone. 😊

Alternative formulation. Alternatively, since $w^T R w$ is already a convex quadratic in the decision variable, we may leave the penalty intact in the objective and apply the epigraph trick only to the peak power. Introducing $\alpha \in \mathbf{R}$ to upper-bound the peak and unfolding the maximum as in (1)–(2) (with $\alpha \geq 0$ implied by feasibility) gives

$$\begin{aligned} & \underset{w, \alpha}{\text{minimize}} && \alpha + \rho w^T R w \\ & \text{subject to} && \|x(t) + S(t)w\|_2^2 \leq \alpha, \quad t = 1, \dots, T. \end{aligned} \quad (8)$$

We expand each squared ℓ_2 norm into a quadratic form, this is, for $t = 1, \dots, T$,

$$\begin{aligned} \|x(t) + S(t)w\|_2^2 &= (x(t) + S(t)w)^T (x(t) + S(t)w) \\ &= w^T S(t)^T S(t) w + 2x(t)^T S(t) w + \|x(t)\|_2^2, \end{aligned}$$

which is a convex quadratic in w , since the Gram matrix $S(t)^T S(t) \geq 0$. Moving α to the left-hand side, each constraint becomes a convex quadratic inequality in (w, α) , yielding

$$\begin{aligned} & \underset{w, \alpha}{\text{minimize}} && \alpha + \rho w^T R w \\ & \text{subject to} && w^T S(t)^T S(t) w + 2x(t)^T S(t) w - \alpha + \|x(t)\|_2^2 \leq 0, \quad t = 1, \dots, T. \end{aligned} \quad (9)$$

The problem in (9) is therefore formulated as a QCQP. 😊

2nd Problem

68. Signal design. Consider a multiple-antenna wireless system with n transmit antennas and m receive antennas. Suppose a transmitted vector $s \in \mathbf{R}^n$ is received as $y \in \mathbf{R}^m$, where

$$y = Hs,$$

with $H \in \mathbf{R}^{m \times n}$ being a given channel matrix. Assume that the magnitude of the transmitter is limited by A , that is, each transmit vector s must obey $\|s\|_\infty \leq A$.

- (a) We want to design a transmit vector s such that the corresponding received vector y resembles a given desired vector y^* , as much as possible in the ℓ_∞ norm:

$$\begin{aligned} & \underset{s}{\text{minimize}} \quad \|Hs - y^*\|_\infty \\ & \text{subject to} \quad \|s\|_\infty \leq A. \end{aligned}$$

Express the problem as a LP, QP, SOCP, or SDP.

- (b) Suppose now that channel matrix H is uncertain. Specifically, suppose that we know only that the entry H_{ij} is in a given interval $[L_{ij}, U_{ij}]$, for each $1 \leq i \leq m$ and $1 \leq j \leq n$. We call

$$\mathcal{H} = \{H \in \mathbf{R}^{m \times n} : L_{ij} \leq H_{ij} \leq U_{ij}, 1 \leq i \leq m, 1 \leq j \leq n\}$$

the uncertainty set for H . To account for the uncertainty in the channel matrix, we adopt a worst-case formulation. We want to design the transmit vector s so as to minimize the maximum deviation that can occur between y and y^* , as H ranges over $\mathcal{H}$:

$$\begin{aligned} & \underset{s}{\text{minimize}} \quad \max \{ \|Hs - y^*\|_\infty : H \in \mathcal{H} \} \\ & \text{subject to} \quad \|s\|_\infty \leq A \end{aligned}$$

Express the problem as a LP, QP, SOCP, or SDP.

Solution to the 2nd Problem. The solution is divided into two parts as follows.

- (a) We apply an epigraph reformulation, introducing a scalar variable $\alpha \in \mathbf{R}$ that upper-bounds the objective, so that the objective becomes linear in the decision variables,

$$\begin{aligned} & \underset{s, \alpha}{\text{minimize}} \quad \alpha \\ & \text{subject to} \quad \|Hs - y^*\|_\infty \leq \alpha, \\ & \quad \quad \quad \|s\|_\infty \leq A. \end{aligned} \tag{10}$$

Feasibility of the first constraint implies $\alpha \geq 0$. Next, we use the fact that, for $z \in \mathbf{R}^d$, $d \in \mathbf{Z}_+$, $\gamma \in \mathbf{R}_+$,

$$\begin{aligned} \|z\|_\infty \leq \gamma & \iff \max\{|z_i| : i = 1, \dots, d\} \leq \gamma \\ & \iff |z_i| \leq \gamma, i = 1, \dots, d \\ & \iff -\gamma \mathbf{1}_d \leq z \leq \gamma \mathbf{1}_d, \end{aligned} \tag{11}$$

which follows from the definition of $\|\cdot\|_\infty$, the fact that a maximum is bounded by γ if and only if every term is (similarly to the argument used in the 1st Problem's solution), and the definition of the absolute value. Above, $\mathbf{1}_d$ denotes the all-ones column vector with d elements.

Therefore, we unfold the constraints in (10) as follows.

- Applying (11) with $z = Hs - y^* \in \mathbf{R}^m$ and $\gamma = \alpha$ gives

$$\|Hs - y^*\|_\infty \leq \alpha \iff -\alpha \mathbf{1}_m \leq Hs - y^* \leq \alpha \mathbf{1}_m \iff \begin{cases} Hs - y^* \leq \alpha \mathbf{1}_m, \\ -Hs + y^* \leq \alpha \mathbf{1}_m; \end{cases} \tag{12}$$

- Applying (11) with $z = s \in \mathbf{R}^n$ and $\gamma = A$ gives

$$\|s\|_\infty \leq A \iff -A \mathbf{1}_n \leq s \leq A \mathbf{1}_n \iff \begin{cases} s \leq A \mathbf{1}_n, \\ -s \leq A \mathbf{1}_n. \end{cases} \tag{13}$$

Substituting these equivalent constraints gives

$$\begin{aligned}
& \underset{s, \alpha}{\text{minimize}} && \alpha \\
& \text{subject to} && Hs - y^\star \leq \alpha \mathbf{1}_m, \\
& && -Hs + y^\star \leq \alpha \mathbf{1}_m, \\
& && s \leq A \mathbf{1}_n, \\
& && -s \leq A \mathbf{1}_n.
\end{aligned} \tag{14}$$

The problem in (14) is therefore an LP, as the objective is linear in the decision variables and the feasible set is a polyhedron. 😊

(b) As before, we start reshaping the problem by introducing an epigraph variable $\alpha \in \mathbf{R}$, so that

$$\begin{aligned}
& \underset{s, \alpha}{\text{minimize}} && \alpha \\
& \text{subject to} && \max \{ \|Hs - y^\star\|_\infty : H \in \mathcal{H} \} \leq \alpha, \\
& && \|s\|_\infty \leq A.
\end{aligned} \tag{15}$$

In this case, the first constraint also depends on H implicitly through the maximum, so we work it out into closed form. To this end, we parametrize the uncertainty set by its easily derived midpoint and radius matrices,

$$\bar{H} := \frac{1}{2}(L + U), \quad R := \frac{1}{2}(U - L),$$

where each entry of R is nonnegative since $U_{ij} \geq L_{ij}$ for $i = 1, \dots, m, j = 1, \dots, n$. Since $\bar{H} - R = L$ and $\bar{H} + R = U$, the constraint $L_{ij} \leq H_{ij} \leq U_{ij}$ can be equivalently written as

$$-R_{ij} \leq H_{ij} - \bar{H}_{ij} \leq R_{ij}, \quad i = 1, \dots, m, \quad j = 1, \dots, n.$$

Therefore, $H \in \mathcal{H}$ if and only if there exists a perturbation matrix $D \in \mathbf{R}^{m \times n}$ such that

$$H = \bar{H} + D, \quad -R_{ij} \leq D_{ij} \leq R_{ij}, \quad i = 1, \dots, m, \quad j = 1, \dots, n.$$

Then, the residual becomes $Hs - y^\star = \bar{H}s - y^\star + Ds$. Before taking the worst case over the uncertainty set, fix an admissible perturbation D . The i -th row of the residual is

$$\sum_{j=1}^n \bar{H}_{ij} s_j - y_i^\star + \sum_{j=1}^n D_{ij} s_j.$$

Therefore, by the definition of the ℓ_∞ norm,

$$\begin{aligned}
\|Hs - y^\star\|_\infty &= \max \left\{ \left| \sum_{j=1}^n H_{ij} s_j - y_i^\star \right| : i = 1, \dots, m \right\} \\
&= \max \left\{ \left| \sum_{j=1}^n \bar{H}_{ij} s_j - y_i^\star + \sum_{j=1}^n D_{ij} s_j \right| : i = 1, \dots, m \right\},
\end{aligned}$$

so the worst-case norm involves two nested maxima, one over the admissible perturbation D and one over the row index i . Since the uncertainty constraints $-R_{ij} \leq D_{ij} \leq R_{ij}$ act on each row of D independently, the choice of D that maximizes row i does not constrain the choice for any other row, and the two maxima commute, that is,

$$\begin{aligned}
& \max \{ \|Hs - y^\star\|_\infty : H \in \mathcal{H} \} \\
&= \max \left\{ \max \left\{ \left| \sum_{j=1}^n \bar{H}_{ij} s_j - y_i^\star + \sum_{j=1}^n D_{ij} s_j \right| : -R_{ij} \leq D_{ij} \leq R_{ij}, j = 1, \dots, n \right\} : i = 1, \dots, m \right\}.
\end{aligned}$$

We therefore freeze the i -th row and solve the inner maximum first. The Triangle Inequality gives

$$\begin{aligned} \left| \sum_{j=1}^n \bar{H}_{ij}s_j - y_i^* + \sum_{j=1}^n D_{ij}s_j \right| &\leq \left| \sum_{j=1}^n \bar{H}_{ij}s_j - y_i^* \right| + \left| \sum_{j=1}^n D_{ij}s_j \right| \\ &\leq \left| \sum_{j=1}^n \bar{H}_{ij}s_j - y_i^* \right| + \sum_{j=1}^n |D_{ij}| |s_j| \\ &\leq \left| \sum_{j=1}^n \bar{H}_{ij}s_j - y_i^* \right| + \sum_{j=1}^n R_{ij} |s_j|, \end{aligned}$$

where the last inequality uses $|D_{ij}| \leq R_{ij}$. The upper bound is clearly achieved by setting each D_{ij} to R_{ij} or $-R_{ij}$ so that $D_{ij}s_j$ has the same sign as the term $\sum_{k=1}^n \bar{H}_{ik}s_k - y_i^*$. Therefore,

$$\max \left\{ \left| \sum_{j=1}^n \bar{H}_{ij}s_j - y_i^* + \sum_{j=1}^n D_{ij}s_j \right| : -R_{ij} \leq D_{ij} \leq R_{ij}, j = 1, \dots, n \right\} = \left| \sum_{j=1}^n \bar{H}_{ij}s_j - y_i^* \right| + \sum_{j=1}^n R_{ij} |s_j|.$$

Substituting back into the initial definition taking the convention that $|s| := (|s_1|, \dots, |s_n|)$, results in

$$\max \{ \|Hs - y^*\|_\infty : H \in \mathcal{H} \} = \|\bar{H}s - y^* + R|s|\|_\infty. \quad (16)$$

Therefore, the constraints in (15) decouple as follows.

- Applying (11) with $z = |\bar{H}s - y^* + R|s| \in \mathbf{R}^m$ and $\gamma = \alpha$, gives

$$\|\bar{H}s - y^* + R|s|\|_\infty \leq \alpha \iff |\bar{H}s - y^* + R|s| \leq \alpha \mathbf{1}_m \iff \begin{cases} \bar{H}s - y^* + R|s| \leq \alpha \mathbf{1}_m, \\ -\bar{H}s + y^* + R|s| \leq \alpha \mathbf{1}_m; \end{cases} \quad (17)$$

- Applying (11) with $z = s \in \mathbf{R}^n$ and $\gamma = A$ gives, as in (a),

$$\|s\|_\infty \leq A \iff \begin{cases} s \leq A \mathbf{1}_n, \\ -s \leq A \mathbf{1}_n. \end{cases} \quad (18)$$

The constraint (17) still contains the absolute value $|s|$. In order to absorb it, we introduce a second epigraph variable $\beta \in \mathbf{R}^n$ such that $|s| \leq \beta \iff -\beta \leq s \leq \beta$. Substituting $R\beta$ for $R|s|$ in (17) and combining with (18) gives the final reformulation of (15), i.e.,

$$\begin{aligned} &\underset{s, \alpha, \beta}{\text{minimize}} && \alpha \\ &\text{subject to} && \bar{H}s + R\beta - y^* \leq \alpha \mathbf{1}_m, \\ & && -\bar{H}s + R\beta + y^* \leq \alpha \mathbf{1}_m, \\ & && s \leq \beta, \\ & && -s \leq \beta, \\ & && s \leq A \mathbf{1}_n, \\ & && -s \leq A \mathbf{1}_n. \end{aligned} \quad (19)$$

As in (a), the problem in (19) is an LP, as the objective is linear in the decision variables and the feasible set is a polyhedron. 😊

3rd Problem

69. Robust linear estimator. A parameter $\theta \in \mathbf{R}$ is observed through a set of n scalar channels, yielding the outputs $y_i = h_i\theta$, for $i = 1, \dots, n$. Here, $h_i \in \mathbf{R}$ is the gain of the i th channel and y_i is the observed output. In vector notation, we have

$$y = h\theta,$$

where $y = (y_1, \dots, y_n)$ and $h = (h_1, \dots, h_n)$.

Our goal is to recover the unknown θ from the available measurement vector $y \in \mathbf{R}^n$. When the channel vector h is known, a popular method consists in estimating θ via linear processing:

$$\hat{\theta} = g^T y$$

where $g = h/\|h\|^2$ (assuming $h \neq 0$). It is straightforward to check that the method works, i.e., $\hat{\theta} = \theta$, for any $\theta \in \mathbf{R}$.

The twist in this problem is that the channel vector h is uncertain. We only know that it lives in a given ellipsoid

$$\mathcal{E}(\bar{h}, P) = \{h \in \mathbf{R}^n : (h - \bar{h})^T P^{-1} (h - \bar{h}) \leq 1\}.$$

We know $\bar{h} \in \mathbf{R}^n$, the center of the ellipsoid, and also $P \in \mathbf{S}_{++}^n$, the matrix that controls the ellipsoid's shape—the eigenvectors of P give the semi-axes of the ellipsoid, and the square-root of the eigenvalues of P give the corresponding lengths (you should sketch the case $\bar{h} = (1, 2)$ and $P = \text{diag}(1, 4)$ to get a feeling for it). For example, if $P = RI_n$ where $R > 0$ and I_n is the $n \times n$ identity matrix, then we know that h lives in a ball of radius $\sqrt{R}$ centered at $\bar{h}$.

A naive method consists in ignoring the uncertainty and apply the popular method pretending that the channel vector h is equal to its nominal value $\bar{h}$. That is, the naive approach builds the estimate $\hat{\theta}_{\text{naive}} = g_{\text{naive}}^T y$, where

$$g_{\text{naive}} = \frac{\bar{h}}{\|\bar{h}\|^2}.$$

In this problem, we develop a linear estimator that takes into account the whole uncertainty region (not only its center, as the naive approach does). We start by noting that, for a fixed $g \in \mathbf{R}^n$, the relative error in the estimate $\hat{\theta} = g^T y$ is

$$\frac{|\hat{\theta} - \theta|}{|\theta|} = |g^T h - 1|$$

and depends on which channel vector h the universe picks from the uncertainty ellipsoid.

A robust formulation seeks the vector g that minimizes the worst relative error, over all possible choices of h , that is,

$$\underset{g}{\text{minimize}} \quad \underbrace{\max \{|g^T h - 1| : h \in \mathcal{E}(\bar{h}, P)\}}_{f(g)}.$$

- Formulate the problem as a LP, QP, QCQP, SOCP or SDP. If you are unable to solve the general case, consider the special case $P = RI_n$. Hint: in either case, your first step should be to find a closed-form expression for the cost function f .
- Solve your formulation via CVX by inserting your code in the appropriate section of the Matlab file `robustestimator.m` (or adapt it to your favorite language). Report your vector g and print the figure generated by the file: it compares the relative errors of both estimators—naive and robust—for a fixed uncertainty ellipsoid and varying θ and h (picked from the ellipsoid).

Solution to the 3rd Problem. The solution is divided into two parts as follows.

- We start by working the cost function into closed form. The set

$$\mathcal{E}(\bar{h}, P) = \{h \in \mathbf{R}^n : (h - \bar{h})^T P^{-1} (h - \bar{h}) \leq 1\}$$

is awkward to maximize over directly because it is defined by a quadratic form in P^{-1} . As in the [1st Problem](#)'s solution, this quadratic form can be written as a squared ℓ_2 norm. Since $P \succ 0$, its eigenvalue decomposition $P = Q\Lambda Q^T$ exists, with Q orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_i > 0$ for $i = 1, \dots, n$, so P is invertible with inverse $P^{-1} = Q\Lambda^{-1}Q^T$. Defining $P^{-1/2} := Q\Lambda^{-1/2}Q^T$, we get $(P^{-1/2})^T = (Q\Lambda^{-1/2}Q^T)^T = Q\Lambda^{-1/2}Q^T = P^{-1/2}$ and $P^{-1/2}P^{-1/2} = P^{-1}$, and so

$$(h - \bar{h})^T P^{-1} (h - \bar{h}) = (h - \bar{h})^T P^{-1/2} P^{-1/2} (h - \bar{h}) = \|P^{-1/2}(h - \bar{h})\|_2^2.$$

The change of variable

$$u := P^{-1/2}(h - \bar{h}) \iff h = \bar{h} + P^{1/2}u$$

therefore reduces the constraint to $\|u\|_2 \leq 1$, transforming $\mathcal{E}(\bar{h}, P)$ into the norm ball

$$\{u \in \mathbf{R}^n : \|u\|_2 \leq 1\},$$

where the inverse map $h = \bar{h} + P^{1/2}u$ uses $P^{1/2} := Q\Lambda^{1/2}Q^T = (P^{-1/2})^{-1}$. Geometrically, as Figure 1 illustrates for the case $\bar{h} = (1, 2)$ and $P = \text{diag}(1, 4)$ from the problem statement, the ellipsoid is just the norm ball stretched by $\sqrt{\lambda_i}$ along each eigenvector q_i of P and translated by $\bar{h}$, and the change of variable inverts this stretching and translation.

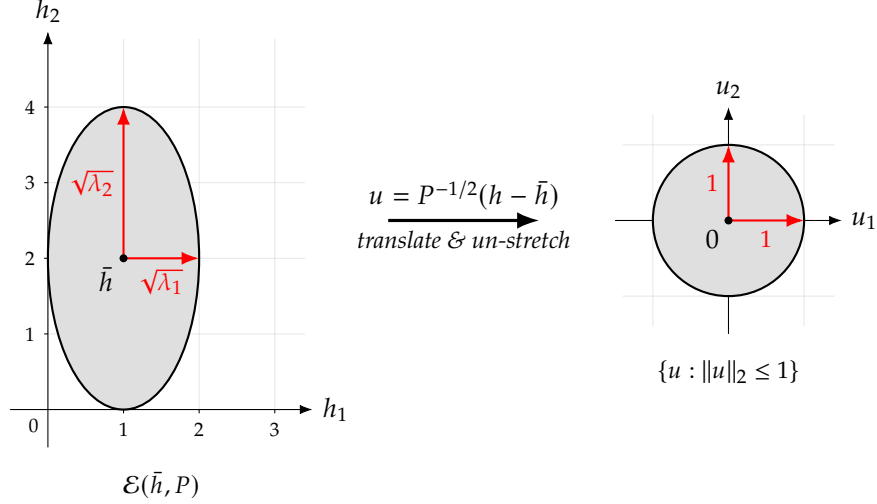


Figure 1: (Left) uncertainty ellipsoid; and (right) the norm ball obtained from the change of variable. With this parametrization, the residual $g^T h - 1$ becomes

$$g^T h - 1 = g^T \bar{h} - 1 + g^T P^{1/2}u = g^T \bar{h} - 1 + (P^{1/2}g)^T u,$$

where we used the symmetry of $P^{1/2}$, namely $(P^{1/2})^T = (Q\Lambda^{1/2}Q^T)^T = Q\Lambda^{1/2}Q^T = P^{1/2}$. The cost function then reads

$$f(g) = \max \{ |g^T \bar{h} - 1 + (P^{1/2}g)^T u| : \|u\|_2 \leq 1 \}.$$

Fixing u with $\|u\|_2 \leq 1$, the Triangle Inequality gives

$$|g^T \bar{h} - 1 + (P^{1/2}g)^T u| \leq |g^T \bar{h} - 1| + |(P^{1/2}g)^T u|,$$

and the Cauchy-Schwarz Inequality, applied to the inner product $(P^{1/2}g)^T u$, gives

$$|(P^{1/2}g)^T u| \leq \|P^{1/2}g\|_2 \|u\|_2 \leq \|P^{1/2}g\|_2,$$

where the last inequality uses $\|u\|_2 \leq 1$. Combining both,

$$|g^T \bar{h} - 1 + (P^{1/2}g)^T u| \leq |g^T \bar{h} - 1| + \|P^{1/2}g\|_2.$$

The upper bound is clearly achieved by setting u to $P^{1/2}g / \|P^{1/2}g\|_2$ or its negative so that $(P^{1/2}g)^T u$ has the same sign as $g^T \bar{h} - 1$. Therefore, the worst-case error admits the closed form

$$f(g) = |g^T \bar{h} - 1| + \|P^{1/2}g\|_2, \quad (20)$$

and the original problem becomes

$$\underset{g}{\text{minimize}} \quad |g^T \bar{h} - 1| + \|P^{1/2}g\|_2.$$

This is not yet in a recognizable form. We now apply an epigraph reformulation to each term, introducing variables $\alpha, \beta \in \mathbf{R}$ that upper-bound the absolute residual and the ℓ_2 norm respectively, so that the objective becomes linear in the decision variables,

$$\begin{aligned} & \underset{g, \alpha, \beta}{\text{minimize}} && \alpha + \beta \\ & \text{subject to} && |g^T \bar{h} - 1| \leq \alpha, \\ & && \|P^{1/2} g\|_2 \leq \beta. \end{aligned} \quad (21)$$

Feasibility of these constraints already implies $\alpha, \beta \geq 0$, since the left-hand sides are nonnegative. Then, the absolute value gives

$$|g^T \bar{h} - 1| \leq \alpha \iff -\alpha \leq g^T \bar{h} - 1 \leq \alpha \iff \begin{cases} g^T \bar{h} - 1 \leq \alpha, \\ -g^T \bar{h} + 1 \leq \alpha, \end{cases}$$

while the constraint $\|P^{1/2} g\|_2 \leq \beta$ is already in standard second-order cone form.

Substituting in (21) gives the final formulation

$$\begin{aligned} & \underset{g, \alpha, \beta}{\text{minimize}} && \alpha + \beta \\ & \text{subject to} && g^T \bar{h} - 1 \leq \alpha, \\ & && -g^T \bar{h} + 1 \leq \alpha, \\ & && \|P^{1/2} g\|_2 \leq \beta. \end{aligned} \quad (22)$$

The problem in (22) is therefore a SOCP, since the objective is linear in the decision variables, the first two constraints are linear inequalities, and the third is a second-order cone constraint. 😊

- (b) The solution, using MATLAB's CVX modeling framework to the final problem formulation in (22) of part (a), is written as follows.

```

12 ...
13 % computes optimal g
14 % insert your code here
15 cvx_begin quiet
16     variable g_robust(n)
17     variable alph
18     variable bet
19
20     minimize( alph + bet )
21
22     subject to
23         g_robust' * h_nom - 1 <= alph;
24         -g_robust' * h_nom + 1 <= alph;
25         norm(A * g_robust, 2) <= bet;
26 cvx_end
27 ...

```

Executing the code results in the optimal robust weight vector

$$g_{\text{robust}}^* \approx (0.0054, 0.1165, 0.0164, 1.0489, 1.0489),$$

with worst-case relative errors

$$\begin{aligned} f(g_{\text{robust}}^*) &= |g_{\text{robust}}^{*T} \bar{h} - 1| + \|P^{1/2} g_{\text{robust}}^*\|_2 \approx 0.1531, \\ f(g_{\text{naive}}) &= \|P^{1/2} g_{\text{naive}}\|_2 \approx 0.7362, \end{aligned}$$

where the absolute term vanishes for the naive estimator, since $g_{\text{naive}}^T \bar{h} = \bar{h}^T \bar{h} / \|\bar{h}\|^2 = 1$.

Figure 2 compares both estimators' performance over 100 Monte Carlo realizations. Evidently, the robust estimator outperforms the naive one, as expected since it minimizes the worst-case error over the entire ellipsoid rather than accounting only for the nominal channel.

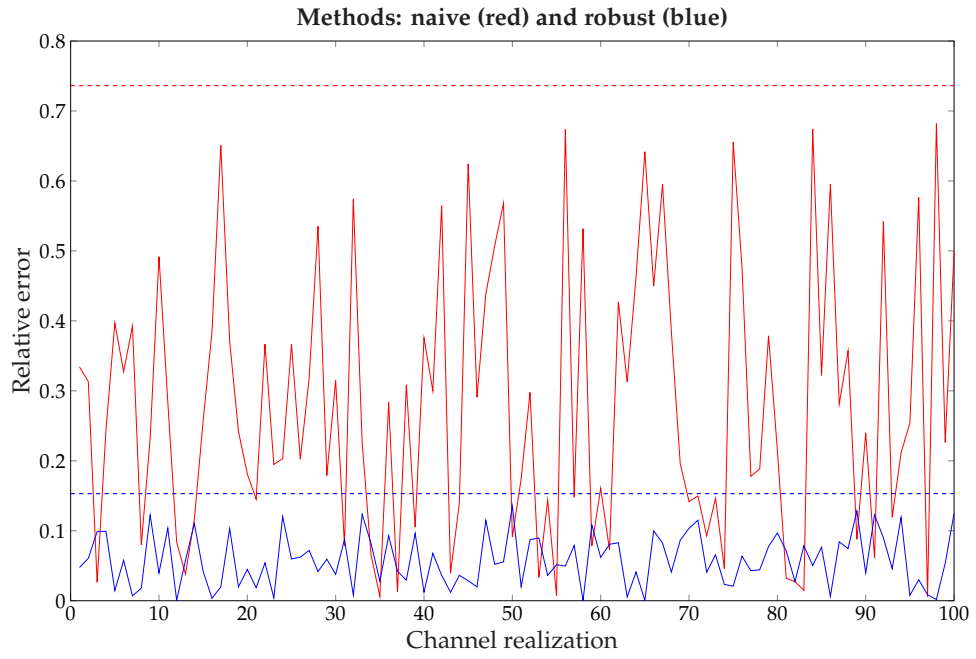


Figure 2: Relative errors of the naive (red) and robust (blue) estimators over 100 Monte Carlo runs; dashed lines of the same color mark the worst-case errors $f(g_{\text{naive}}) \approx 0.7362$ and $f(g_{\text{robust}}^{\star}) \approx 0.1531$.